8.6 Completing the Square When $a > 1$ Using Algebra Tiles

Lesson Introduction

Benchmarks	MA.912.AR.3.6 Given an expression or equation representing a quadratic function, determine the vertex and zeros and interpret them in terms of a real-world context. MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.	Learning Target	• I can use algebra tiles to complete the square for a quadratic expression where $a > 1$.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Question	• How can algebra tiles help me demonstrate the process of completing the square when $a > 1$?
Lesson Narrative	In this lesson, students extend their learning of completing the square to algebraic expressions where $a > 1$. Students will use algebra tiles to explore this concept and use that experience to make conjectures about the process of completing the square. They will learn the formalized process for completing the square when $a > 1$.

Component Summary	8.6.1 Warm-Up (~5 min.)	Bell Work related to the understanding of completing the square for a quadratic expression where $a = 1$ *(SE p. 38)*	8.6.3 Bring It Together (15 min.)	Work to connect the use of algebra tiles to complete the square and the algebraic equivalence *(SE p. 41)*
	8.6.2 Exploration (20 min.)	Explore the concept of completing the square where the value of $a > 1$ using algebra tiles *(SE p. 39)*	8.6.4 Wrap-Up (~5 min.)	Complete a reflection activity to show the level of understanding of the current materials *(SE p. 42)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Completing the Square	• (Optional) Algebra Tiles	• N/A

Teacher Tips

Common Misconceptions

- Students may struggle understanding why there are "a" number of mats, missing the connection between the leading coefficient and number of mats.
- Students may inadvertently add negative 1 tiles when working with negative x tiles.

Things to Consider

- Students began using algebra tiles in the prior lesson to connect to the concept of completing the square. In the prior lesson, the expressions were limited to an a value of 1.
- Consider engaging with 8.6.2 Exploration as a student in order to identify possible misconceptions or teaching moments to leverage specifically to your students based on their progress from Lesson 5.

Literacy and Language Routines

Compare and Connect

8.6.3 Bring It Together brings together the concept of completing the square with algebra tiles where the value of $a > 1$ with the algebraic equivalent. In the process of bringing together those concepts, consider demonstrating your thinking out loud and prompting students to reflect and respond.

Mathematical Thinking and Reasoning (MTR) Standard

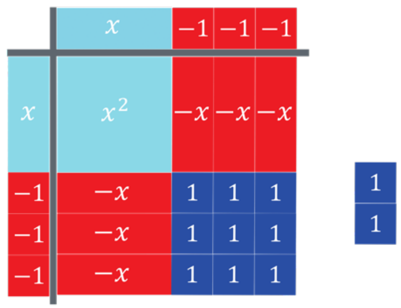
MA.K12.MTR.2.1: Multiple Representations

8.6.3 Bring It Together has students connect their method of using algebra tiles to an algebraic approach. While students are not using the algebraic approach in this lesson, this activity draws the connection to set students up for using this method in the next lesson.

Differentiate Instruction

Algebra Tiles

Algebra tiles drive learning and instruction during this lesson. Students should have access to algebra tiles and a product map, either individually or in pairs, for access to conceptual understanding during each activity. If resources are limited, consider having students create algebra tiles with strips of blank paper, markers, and scissors or virtual algebra tiles. Additionally, construct one station accessible to all students in class that they can visit with a set of algebra tiles to use as an exemplar for making their own as well as scaffolding to use during activities in the lesson.



Lesson Notes

8.6.1 Warm-Up: Bell Work

Component Overview: *Students will complete Bell Work related to their understanding of completing the square for a quadratic expression where $a = 1$.*

8.6.1 Warm-Up: Bell Work

1. Use the diagram to complete the square for the expression, $x^2 + 10x + 8$.

	x	1	1	1	1	1
x	x^2	x	x	x	x	x
1	x	1	1	1	1	1
1	x	1	1	1		
1	x					
1	x					
1	x					

$(x+5)^2 - 17$

Consider pairing student responses from 8.5.4 Wrap-Up with answers from this activity to inform differentiated groups or instruction throughout this lesson.

- *Students may add 17 instead of subtracting 17. Students may be unable to algebraically produce their understanding of counting missing tiles.*

8.6.2 Exploration: Completing the Square Where $a > 1$ Using Algebra Tiles

Component Overview: *Students will explore the concept of completing the square where the value of $a > 1$ using algebra tiles.*

8.6.2 Exploration: Completing the Square Where $a > 1$ Using Algebra Tiles

1. The diagram shows three perfect square trinomials.

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

Complete each representation of the diagram.

Three Perfect Square Trinomials	Three Binomials Squared	All Like Terms Added
$3(x^2 + 6x + 9)$	$3(x + 3)^2$	$3x^2 + 18x + 27$

For this activity, keep in mind that the structure is intended to be an exploration. If necessary, question 1 can be used as an instructional lead or "we do" to model thinking and exploration.

Consider leveraging the 8.5.4 Wrap-Up and 8.6.1 Warm-Up to inform student pairing or groups.

8.6.2 Exploration: Completing the Square Where $a > 1$ Using Algebra Tiles (continued)

2. The expression $3(x^2 + 6x) + 11$ is represented in the diagram using three product mats.

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	1
1	x	1	1	1
1	x	1	1	1

	x	1	1	1
x	x^2	x	x	x
1	x	1	1	
1	x			
1	x			

	x	1	1	1
x	x^2	x	x	x
1	x			
1	x			
1	x			

a. Discuss this representation with your partner.
Answers will vary. Each square has the same number of x^2's and x's but different unit tiles.

For question 2a, why do you think there are two tile mats? Would it be possible to make a square using two x^2 tiles?

b. Name the property of operations that is applied to rewrite $3x^2 + 18x + 11$ as $3(x^2 + 6x) + 11$.
Distributive Property.

c. How many unit tiles (1) should be added to complete all of the squares?
16 *tiles*

For the unit tiles in question 2c, how do you know if those unit tiles will be positive or negative?

By adding those unit tiles, the diagram now shows $3(x + 3)^2$. As a reminder, if 1-unit tiles are added to the model, then the same number of -1-unit tiles must be added to the model so that its value doesn't change. Algebraically, this can be represented by subtracting that number of units.

For an added layer, consider having students explain how they determined their answer in question 2d. Additionally, you can place limitations on how they do that, such as with algebra tiles or algebraically.

d. Complete the expression.

$3(x + 3)^2 - 16$

3. The diagram of algebra tiles shown represents the expression $2x^2 - 16x - 5$.

	x	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$
-1	$-x$				
-1	$-x$				
-1	$-x$				
-1	$-x$				

	x	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$
-1	$-x$				
-1	$-x$				
-1	$-x$				
-1	$-x$				

-1	-1
-1	-1
-1	

a. Rewrite the expression $2x^2 - 16x - 5$ to show the representation of the two separate squares by factoring 2 out from the first two terms.

$2(x^2 - 8x) - 5$

b. How many unit tiles (1) should be added to complete **one** of the squares?
16 tiles

c. Add the number of unit tiles for **one** square to the expression inside the parentheses.

$2(x^2 - 8x + 16) - 5$

d. Describe how the coefficient of x is related to the number of unit tiles added for **one** square.

$2(x^2 - 8x + \_\_\_\_\_) - 5$

Answers will vary. The number of unit tiles is half the value of the coefficient of x then squared.

Watch for student misconceptions in question 3c. Students may inadvertently misread the instructions and add the 16 to the value of -5 and not to the expression inside of the parentheses.

Emphasize questions 3d, 3e, and 3g when reviewing the activity with students. Prompt them to express their responses to these parts in their own words, as their answers to questions 3d, 3e, and 3g move understanding from pictorial to conceptual.

8.6.2 Exploration: Completing the Square Where $a > 1$ Using Algebra Tiles (continued)

e. Discuss with your partner how the 2 outside of the parentheses is related to the diagram.
Answers will vary. The are two squares.

f. What is the total number of unit tiles added to complete the squares?
32 *unit tiles*

g. Discuss with your partner how multiplying the 2 outside of the parentheses to the number of unit tiles added for **one** square also gives the total number of unit tiles added.

$$2(x^2 - 8x + \underline{\qquad}) - 5$$

Answers will vary. It applies the distributive property to ensure the number of one squares represents the entire expression.

h. Complete the expression to show the subtraction of the total number of unit tiles so the new expression remains equivalent to the original expression, $2x^2 - 16x - 5$.

$$2(x^2 - 8x + 16) - 5 - 32$$

i. Use the completed squares and the added tiles to rewrite the expression $2x^2 - 16x - 5$.
$2(x - 4)^2 - 37$

Consider inquiry to drive exploration.

- *Since each square requires 16 tiles to complete the square, where is this represented in the algebraic expression?*
- *How does the subtraction of 32 relate to zero pairs?*
- *How can you check that the expression is equivalent to the original expression?*

8.6.3 Bring It Together: Completing the Square Where $a > 1$ Using Algebra Tiles

Component Overview: *Students work to connect their use of algebra tiles to complete the square and the algebraic equivalence. For this activity, walk students through the steps of this activity.*

8.6.3 Bring It Together: Completing the Square Where $a > 1$ Using Algebra Tiles

The diagram of algebra tiles shown represents the expression $4x^2 - 48x - 3$.

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

	x	-1	-1	-1	-1	-1	-1
x	x^2	$-x$	$-x$	$-x$	$-x$	$-x$	$-x$
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						
-1	$-x$						

$\boxed{-1}$

$\boxed{-1}$

$\boxed{-1}$

As this activity begins, students may be confused on why there are four product mats on this example. Connect the four product mats to the coefficient of x^2. Help students see the representation of each value $4, 48,$ and -3.

If you have enough materials, consider providing students with the product mats and algebra tiles so they can bring together their learning throughout this activity.

8.6.3 Bring It Together: Completing the Square Where $a > 1$ Using Algebra Tiles (continued)

1. Fill in the table to complete the square.

Process	Expression
Original Expression	$4x^2 - 48x - 3$
Factor the first two terms to reflect the algebra tiles in each square. The number factored out is also the **number of squares**.	$4(x^2 - 12x) - 3$
Determine the number of **unit tiles** that should be added for each square.	$4(x^2 - 12x + 36) - 3$
Multiply the **number of squares** by the number of **unit tiles** added to each square. Subtract the **total number of unit tiles** added.	$4(x^2 - 12x + 36) - 3 + (-144)$
Rewrite the perfect square trinomial represented in the square as a binomial squared.	$4(x - 6)^2 - 3 + (-144)$
Combine like terms.	$4(x - 6)^2 + (-147)$

Consider using a Foldable, note-taking guide, or Anchor Chart to create a reference document for students that display this process for students to refer back to as they move forward in additional units throughout the remainder of this unit.

Consider using an inquiry-based approach to position students as sense makers.

- *Why do you think that the value that is factored out is equal to the number of squares?*
- *How is the -12 shown on the algebra tiles?*
- *How is the 36 shown on the algebra tiles?*
- *Why is -144 added?*
- *How does the process show what was done using algebra tiles?*

8.6.4 Wrap-Up: Reflection

***Component Overview:** Students will complete a reflection activity to show their level of understanding of the current materials. This activity should be done independently to provide appropriate student-level data.*

8.6.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding.

   Answers will vary.

   I'm so confident, I could explain this to someone else!

   I can get to the right answer, but I don't understand well enough to explain it yet.

   I understand some of this, but I don't understand all of it yet.

   I tried hard, and I listened, but I am finding this challenging. I will make sure that I get help.

   I do not understand any of this yet. I listed my questions below.

This provides data for how students feel about their level of understanding of the content. Couple this data with student performance in this lesson to determine which students may need additional support with specific key features.